SHENGHONG WANG

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Higher Education

Doctor of Philosophy (Ph.D.) in Environmental Sciences & Health Sep. 2020 – Aug. 2024
University of Nevada, Reno Reno, NV, USA

Honours Bachelor of Science in Environmental Physics Sept. 2015 – Jun. 2019
University of Toronto Toronto, ON, Canada

Professional Experience

Physicians Committee for Responsible Medicine Remote in Tomball, TX
Regulatory Testing Specialist Feb. 2026 - Current

- Reduce and replace the use of animals in toxicology by leading the development, evaluation, and promotion of human-relevant in vitro and computational approaches for chemical safety assessment across multiple toxicological endpoints.

Dow Chemical Company Midland, MI
Product Safety Research Cheminfomatics Postdoctoral Fellow Jan. 2025 – Jan 2026

- Developed KNIME workflows to evaluate QSAR models for mutagenicity predictions, improving insights into each model's applicability domain and performance for chemical risk assessment
- Compiled a database integrating mutagenicity and potential mode of action to enhance the understanding of chemicals structure and mutagenicity relationship

Arnot Research & Consulting Houston, TX
Research Associate Sep. 2025 – Jan. 2025

- Conduct high-throughput toxicokinetics (HTTK) modeling on over 70,000 organic chemicals, demonstrating significant differences between various approaches and underscoring the need for imporving HTTK methods, especially when using New Approach Methodologies (NAMs) for chemical screening

University of Nevada, Reno Reno, NV
Research Assistant Jan. 2021 – Aug. 2024

- Lead the in-vitro to in-vivo extrapolation (IVIVE) modeling for first-pass metabolism parameters, enhancing the understanding of human in-vivo metabolism through in-vitro assay
- Develop a novel oral absorption model, significantly expanding the applicability domain and predictive power of existing models to better understand internal exposure in the context of NAMs
- Create a mass balance membrane permeability model, highlighting the limitations of current in-vitro designs and informing the applicability domain of in-vitro transmembrane permeability assays, improving IVIVE efforts for chemical absorption
- Initiate and lead the student team as Principal Investigator (PI) for the U.S. Environmental Protection Agency People, Prosperity, and the Planet (P3) grant application to develop computational methods to predict how factors such as social-economic, human behaviors, and physiology can affect the chemical concentrations among the U.S. populations

University of Nevada, Reno Reno, NV
Teaching Assistant Jan. 2021 – Sep. 2022

- Lead tutorial, provided individual assistance, and prepared educational materials for the environmental exposure assessment course focused on the processes and factors affecting human exposure to environmental and occupational contaminants.

Publications

Chen, Z., Wang, S., Li, L., Xia, K., Wang, X., Gong, Y., & Liu, Q. (2026). Heterogeneous ozone oxidation of p-phenylenediamine antioxidants. *Journal of Hazardous Materials*, *507*, 141745.

Fahy, W. D., Zhang, Z., Wang, S., Li, L., & Mabury, S. A. (2025). Environmental fate of the azole fungicide fluconazole and its persistent and mobile transformation product 1, 2, 4-triazole. *Environmental Science & Technology*, *59*(6), 3239–3251.

Illa, S. E., Li, Y., Wang, S., Li, D., & Li, L. (2025). Systematic evaluation of in vitro intestinal permeability measurements across diverse chemicals: Insights for advancing new approach methods for chemical risk assessment. *Environmental Science & Technology*, *59*(48), 25575–25586.

Liu, M., Wang, S., Chen, X., Wang, M., Zhao, H., Zhao, F., Cui, S., Li, L., & Fang, M. (2025). Separating the role of gut enzymatic transformation in modulating internal exposure to three major phthalates. *Environmental Science & Technology*, *59*(16), 7867–7876.

Zhang, Z., Wang, S., & Li, L. (2025). Insights from modeling the fate of and exposure to persistent and mobile organic chemicals using the refined protex model. *Environmental Toxicology and Chemistry*, *44*(11), 3349–3362.

Fahy, W. D., Gong, Y., Wang, S., Zhang, Z., Li, L., Peng, H., & Abbatt, J. P. (2024). Hydroxyl radical oxidation of chemical contaminants on indoor surfaces and dust. *Proceedings of the National Academy of Sciences*, *121*(45), e2414762121.

Intasiri, A., Illa, S. E., Prertprawnon, S., Wang, S., Li, L., Bell, T. W., & Li, D. (2024). Comparison of in vitro membrane permeabilities of diverse environmental chemicals with in silico predictions. *Science of the Total Environment*, *933*, 173244.

Wang, J., Wang, S., Zhang, Z., Wang, X., Xia, K., Li, L., & Liu, Q. (2024). Understanding the importance of atmospheric transformation in assessing the hazards of liquid crystal monomers. *Environmental Science: Processes & Impacts*, *26*(1), 94–104.

Wang, S., Zhang, Z., Li, D., & Li, L. (2024). Elucidating impacts of partitioning and transmembrane permeability on absorption of chemicals in human gastrointestinal tract. *Environment international*, *193*, 109108.

Wang, S., Zhang, Z., Saunders, L. J., Li, D., & Li, L. (2024). Understanding the impacts of presystemic metabolism on the human oral bioavailability of chemicals. *Environmental Science & Technology*, *58*(32), 14135–14145.

Zhang, Z., Sangion, A., Wang, S., Gouin, T., Brown, T., Arnot, J. A., & Li, L. (2024). Chemical space covered by applicability domains of quantitative structure–property relationships and semiempirical relationships in chemical assessments. *Environmental Science & Technology*, *58*(7), 3386–3398.

Zhang, Z., Wang, S., Brown, T. N., Sangion, A., Arnot, J. A., & Li, L. (2024). Modeling sorption of environmental organic chemicals from water to soils. *Water Research X*, *22*, 100219.

Zhang, Z., Sangion, A., Wang, S., Gouin, T., Brown, T., Arnot, J. A., & Li, L. (2023). Hazard vs. exposure: Does it make a difference in identifying chemicals with persistence and mobility concerns? *Water Research*, *245*, 120610.

Li, L., Zhang, Z., Men, Y., Baskaran, S., Sangion, A., Wang, S., Arnot, J. A., & Wania, F. (2022). Retrieval, selection, and evaluation of chemical property data for assessments of chemical emissions, fate, hazard, exposure, and risks. *ACS Environmental Au*, *2*(5), 376–395.

Wang, S., Zhang, Z., Li, D., Illa, S. E., & Li, L. (2022). In silico model-based exploration of the applicability of parallel artificial membrane permeability assay (pampa) to screen chemicals of environmental concern. *Environment International*, *170*, 107589.

Zhang, Z., Wang, S., & Li, L. (2021). Emerging investigator series: The role of chemical properties in human exposure to environmental chemicals. *Environmental Science: Processes & Impacts*, *23*(12), 1839–1862.

Presentations

Select Platform Presentations

- Introduction to PBPK Modeling and Pharmacokinetic Principles. 2026 Summer Immersion on Innovative Approaches in Science, Baltimore, Jun. 15-18, 2026
- In-Silico Model-based Exploration of the Importance of Gut Metabolism in Human Exposure and Toxicokinetic Modeling. ACS FALL 2023, San Francisco, August 13 - 17, 2023
- In-Silico Model-based Exploration of the Importance of Gut Metabolism in Human Exposure and Toxicokinetic Modeling. Society of Environmental Toxicology and Chemistry 43rd North American Annual Meeting. Pittsburgh, Nov. 13–17, 2022

- In-silico PAMPA Model: Considerations for Applying PAMPA to Environmental Chemicals. Society of Environmental Toxicology and Chemistry 42nd North American Annual Meeting. (Online). Nov. 14–18, 2021

Select Poster Presentations

- In-Silico Model-based Exploration of the Importance of Gut Metabolism in Human Exposure and Toxicokinetic Modeling. SETAC Europe 33rd Annual Meeting. Dublin. Apr.30 – May 4, 2023
- In-silico PAMPA Model: Considerations for Applying PAMPA to Environmental Chemicals. SETAC Europe 33rd Annual Meeting. Dublin. Apr.30 – May 4, 2023
- In-Silico Model-based Exploration of the Importance of Gut Metabolism in Human Exposure and Toxicokinetic Modeling. Society of Toxicology 62st Annual Meeting and ToxExpo. Nashville, TN. Mar. 19–23, 2023
- In-silico PAMPA Model: Considerations for Applying PAMPA to Environmental Chemicals. Society of Toxicology 62st Annual Meeting and ToxExpo. Nashville, TN. Mar. 19–23, 2023
- In-silico PAMPA Model: Considerations for Applying PAMPA to Environmental Chemicals. Society of Environmental Toxicology and Chemistry 43rd North American Annual Meeting. Nov. 13–17, 2022
- In-silico PAMPA Model: Considerations for Applying PAMPA to Environmental Chemicals. Society of Toxicology 61st Annual Meeting and ToxExpo. San Diego. CA, United States. Mar. 27–31, 2022
- In-silico PAMPA Model: Considerations for Applying PAMPA to Environmental Chemicals. Society of Environmental Toxicology and Chemistry 32nd Europe Annual Meeting. (Online). May 15–19, 2021

GRANTS

U.S. Environmental Protection Agency People, Prosperity and the Planet (P3), $75,000

- Student PI, responsible for study design, organizing research team, and writing grant proposal

U.S. Environmental Protection Agency (EPA-G2019-STAR-D1)

- Main participant

HONORS, SCHOLARSHIPS AND AWARDS

Honors

- Being featured on the cover page of Environmental Science: Processes Impacts (ESPI), Zhang, Z., Wang, S., Li, L. (2021). Emerging investigator series: the role of chemical properties in human exposure to environmental chemicals. Environ. Sci.: Process. Impacts, 23 (12), 1839-1862
- Being featured as ESPI's 2021 best papers, Zhang, Z., Wang, S., Li, L. (2021). Emerging investigator series: the role of chemical properties in human exposure to environmental chemicals. Environ. Sci.: Process. Impacts, 23 (12), 1839-1862
- Being featured on the cover page of Environ. Sci. Technol (ES&T), Zhang,Z., Sangion, A., Wang, S., Gouin, T., Brown, T., Arnot, JA., Li, L. (2023) Chemical Space Covered by Applicability Domains of Quantitative Structure-Property Relationships and Semi-empirical Relationships in Chemical Assessments. Environ. Sci. Technol.
- Being featured on the cover page of Environ. Sci. Technol (ES&T), Illa, S. E., Li, Y., Wang, Shenghong, Li, D., Li, L. (2025). Systematic evaluation of in vitro intestinal permeability measurements across diverse chemicals: Insights for advancing new approach methods for chemical risk assessment. Environ. Sci. Technol.

Scholarships

- Full scholarship to support Ph.D. studies in Environmental Science, including tuition, stipend, and research support by U.S. Environmental Protection Agency EPA-G2019-STAR-D1
- UNR Institutional Methodology award (2023 to 2024)

Awards

- University of Nevada, Reno Student Development Award, $2,000
- University of Nevada, Reno Graduate Student Travel Award, $500
- 2023 Society of Toxicology (SOT) Annual Meeting, Biological Modeling Speciality Section, Best Trainee Abstract finalist
- 2024 SOT Annual Meeting, Biological Modeling Speciality Section, Best Trainee Abstract finalist

Academic Service

Session Chair

- 2026 Summer Immersion on Innovative Approaches in Science Introduction to PBPK Modeling and Pharmacokinetic Principles   Baltimore, MD, Jun. 2026
- American Society for Cellular and Computational Toxicology (ASCCT) Continuous Education course on PBTK Modeling: Principles, Applications, and Hands-On Practice   Durham, NC, Oct 2026

Professional Affiliations / Memberships

- Member of Society of Environmental Toxicology and Chemistry 2021 - 2023
- Member of Society of Toxicology 2025-current
- Member of American Society for Cellular and Computational Toxicology 2026-current

Peer-Reviewer (Journals)

- Environmental Health Perspectives
- Environment International
- Environmental Pollution